

#

Lens

- Lens

↳ Lens is a portion of transparent refracting medium bounded by one or two curved surfaces.

- Conver lens

↳ The lens which is thick at the middle and thin at the edges is called conver lens.

It is also called Converging lens because it converges a parallel beam of light falling on it to a point.

- Concave lens

↳ The lens which is thin at the middle and thick at the edges is called concave lens.

It is also called diverging lens because it diverges a parallel beam of light falling on it from a point.

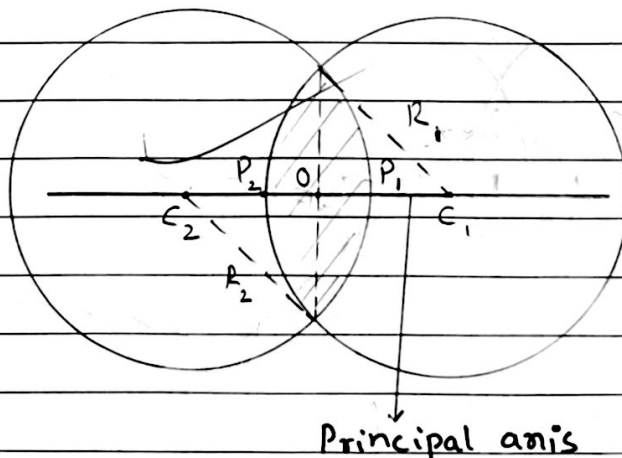


fig- Conver lens

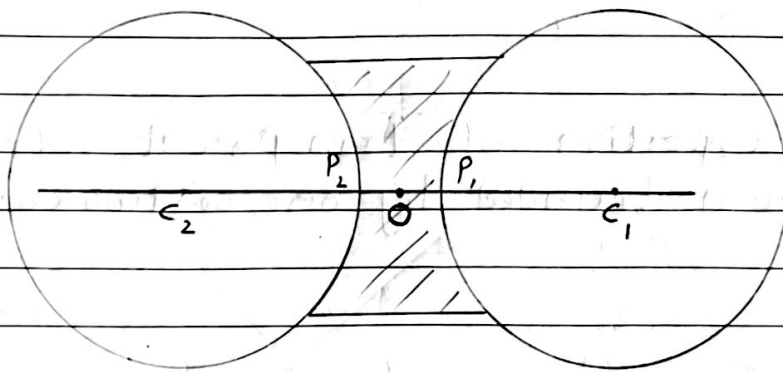


fig- Concave lens

- Optical centre (O)
 ↳ It is the centre of the lens from which rays of light will pass straightly without bending.
- Optical axis
 ↳ It is a line perpendicular to principal axis and passes from optical centre.
- Rules to draw ray diagram.

①

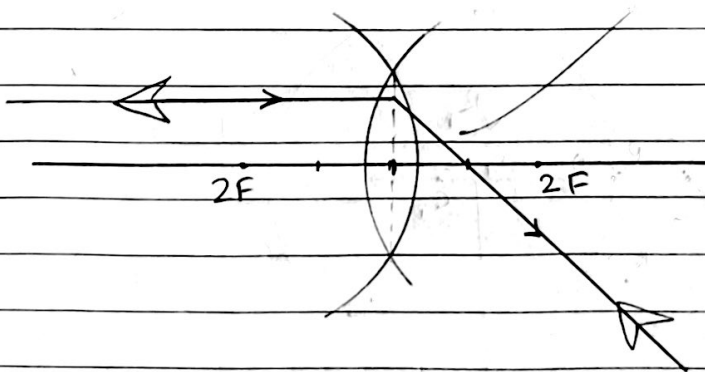


fig- Convex lens.

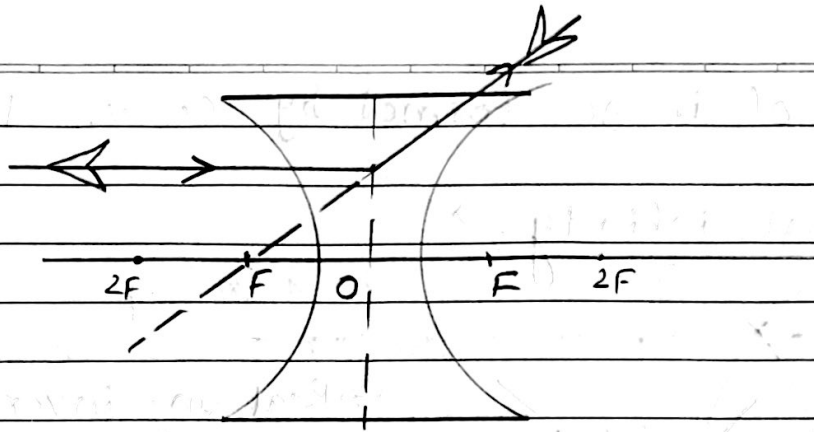


fig - Concave lens

(2)

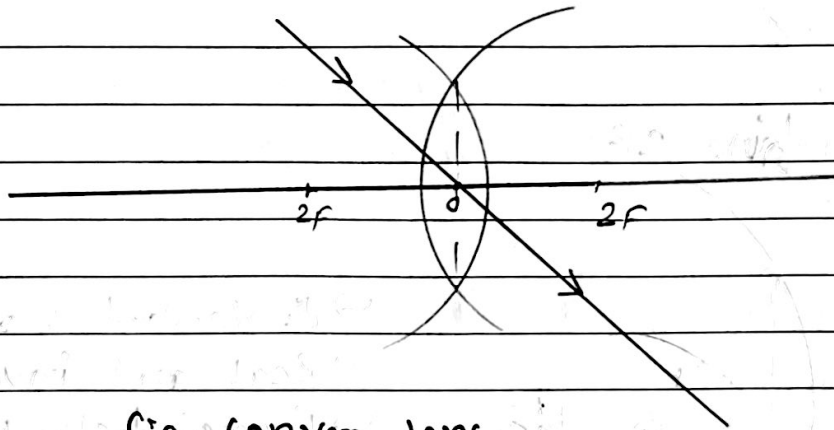


fig - convex lens.

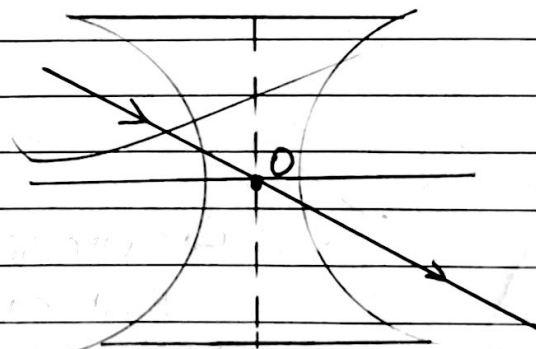
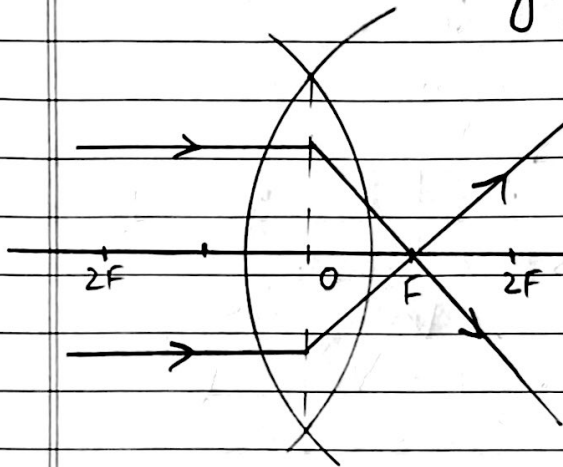


fig - concave lens.

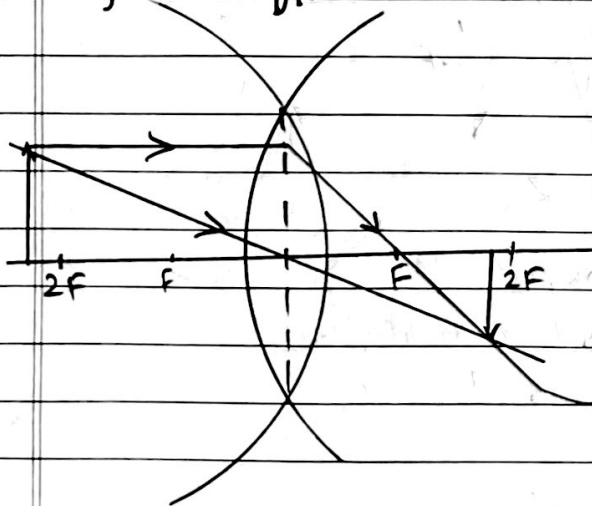
• Nature of image formed by convex lens

① Object at infinity.



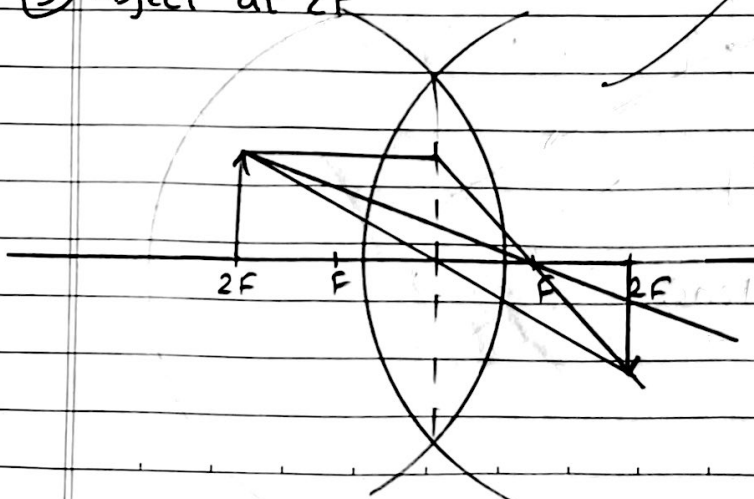
- ↳ Real and inverted
- ↳ Highly diminished
- ↳ Formed at F on opposite side

② object behind $2F$



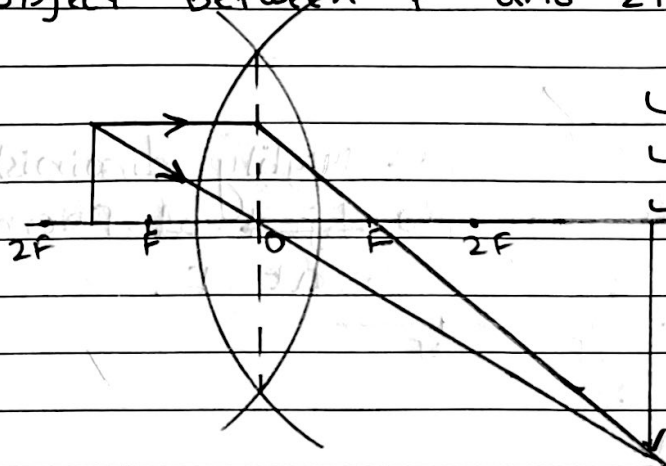
- ↳ diminished ~~and real~~
- ↳ Real and inverted
- ↳ Formed between F and $2F$ on opposite side.

③ Object at $2F$



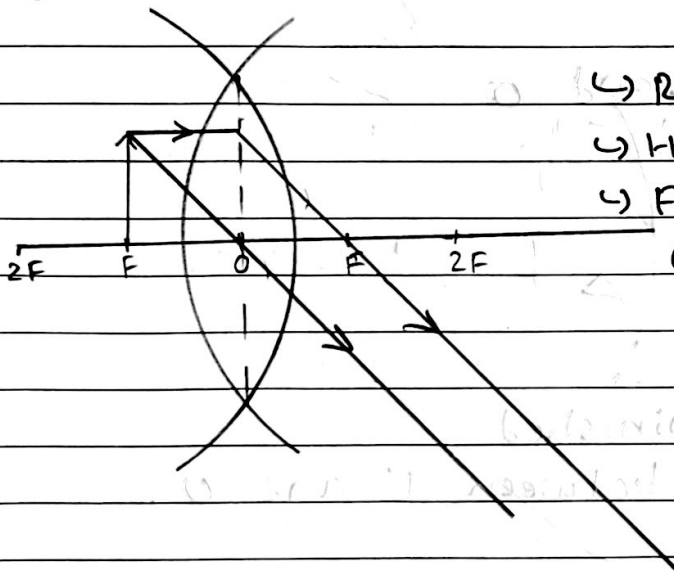
- ↳ Same size ~~image~~
- ↳ Real and inverted
- ↳ Formed at $2F$ on opposite side

④ Object between F and $2F$



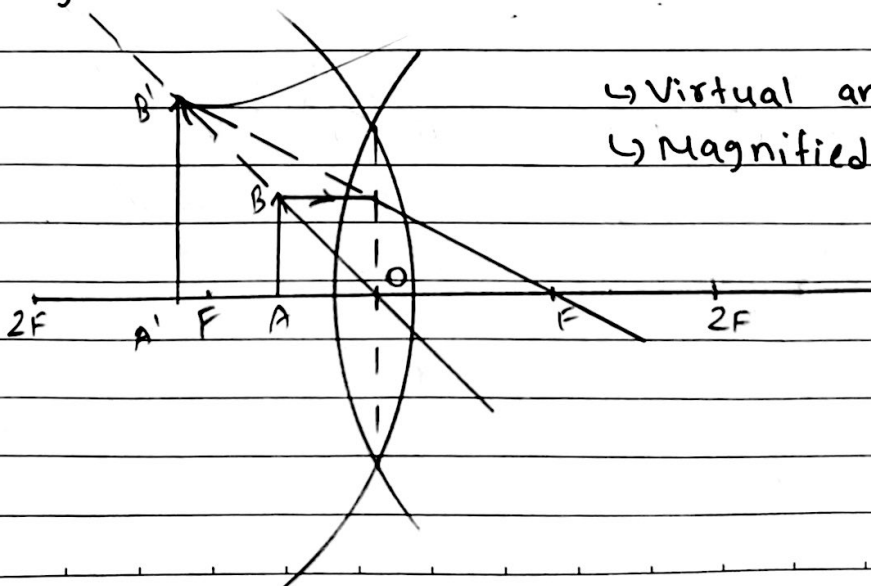
- ↳ Real and inverted
- ↳ Magnified
- ↳ Formed behind $2F$ on opposite side.

⑤ Object at F



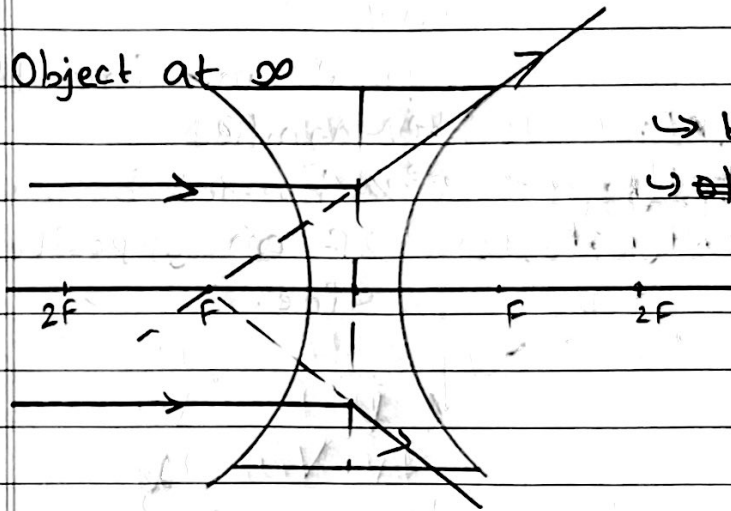
- ↳ Real and inverted
- ↳ Highly Magnified
- ↳ Formed at infinity on opposite side.

⑥ Object between F and O



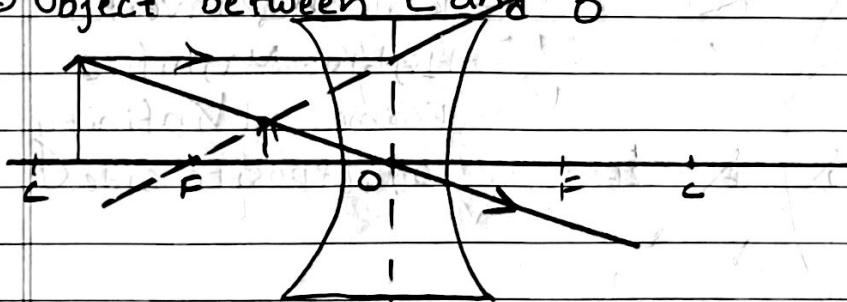
- ↳ Virtual and erect
- ↳ Magnified

① Object at ∞



→ Highly diminished
→ ~~Object~~ at F formed
at F .

② Object between C and O



→ ~~Obj.~~ Diminished
→ Formed between F and O .

lens formula $\left[\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \right]$

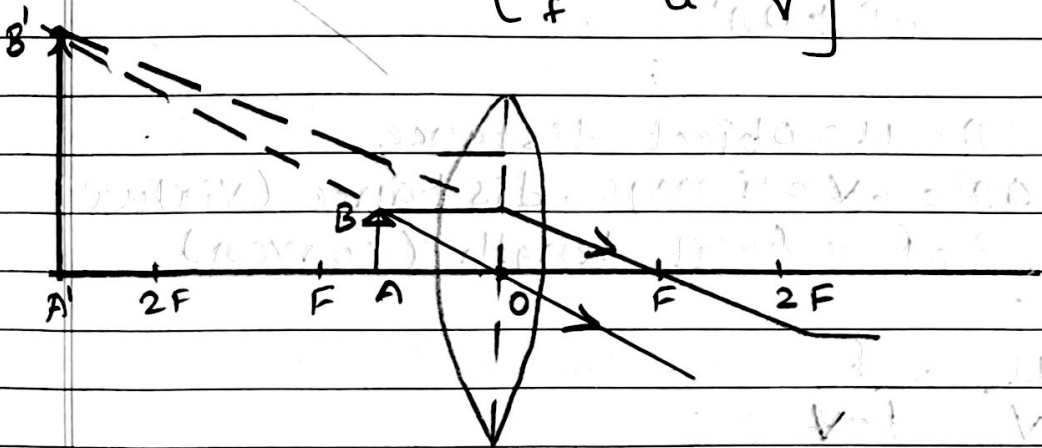


fig: Convex lens when virtual image is formed

Consider a convex lens of focal length f . AB is the object beyond F and $A'B'$ is the image of that object.

Now, From Figure,

$$\triangle AOB \sim \triangle A'O B'$$

$$\therefore \frac{AB}{A'B'} = \frac{OA}{OA'} \rightarrow \textcircled{i}$$

$$\text{Also, } \triangle ODF \sim \triangle A'B'F$$

$$\therefore \frac{OD}{A'B'} = \frac{OF}{A'F}$$

$$\Rightarrow \frac{AB}{A'B'} = \frac{OF}{A'F} \rightarrow \textcircled{ii} \quad [\because AB = OD]$$

From eqn $\textcircled{i}$ and $\textcircled{ii}$

$$\frac{OA}{OA'} = \frac{OF}{A'F}$$

$$\Rightarrow \frac{OA}{OA'} = \frac{OF}{OF + OA'}$$

Here, $OA = u =$ object distance

$OA' = -v =$ image distance (Virtual)

$OF = f =$ focal length (convex)

$$\therefore \frac{u}{-v} = \frac{f}{f-v}$$

$$\text{or, } uf - uv = -vf$$

$$\text{or, } \frac{uf}{uf} - \frac{uv}{uf} = \frac{-vf}{uf}$$

$$\Rightarrow \frac{1}{u} - \frac{1}{v} = \frac{-1}{f}$$

$$\Rightarrow \frac{1}{u} - \frac{1}{v} = -\frac{1}{f}$$

$$\therefore \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Magnification

↳ It is the ratio of size of the image to the size of the object.

$$\text{i.e. } m = \frac{v}{u} = \frac{I}{O} = \frac{\text{size of image (I)}}{\text{size of object (O)}}$$

From eqn (i)

$$\frac{AB}{A'B'} = \frac{OA}{OA'}$$

$$\text{or, } \frac{O}{-I} = \frac{u}{-v}$$

$$\Rightarrow \frac{I}{O} = \frac{V}{u}$$

$$m = \frac{v}{u}$$

- Lens Maker's formula $\left[\frac{1}{f} = (n-1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \right]$

→ lens maker's formula is a relation which connects the focal length of a lens to the radii of curvature of the two surfaces of the lens and the refractive index of the material of the lens.

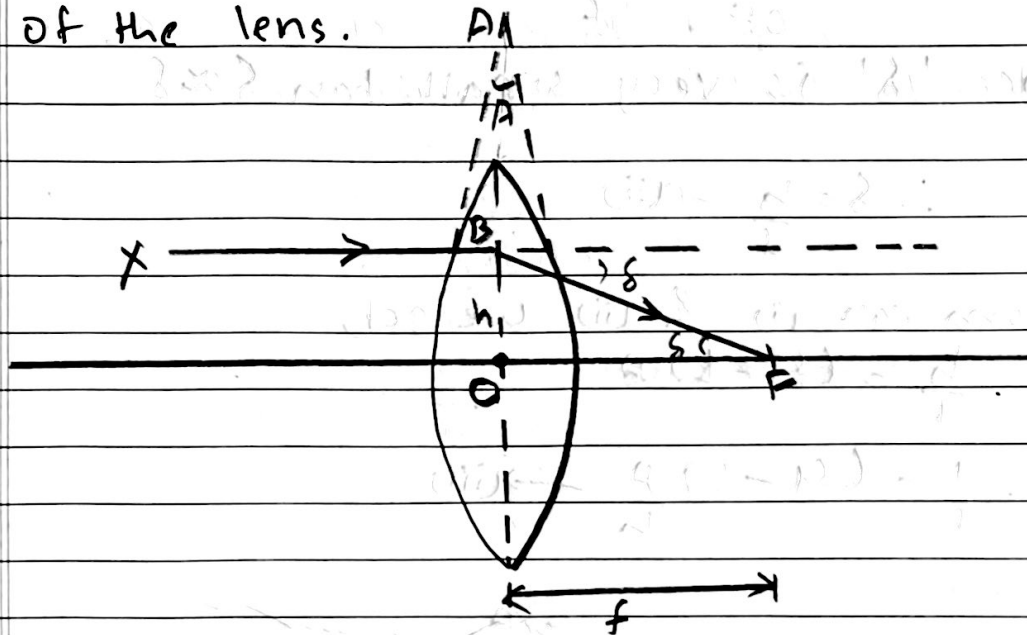


fig. Refraction through lens.

Consider a thin convex lens of focal length 'f'. A ray of light X-B parallel to the principal axis strikes the point B at height 'h' above the optical centre O as shown in fig. This ray after ref-

ray passing through lens meets principal axis at focus making angle of deviation ' δ '. Refracting angle of prism (A) is formed drawing tangents at the light deviating points of the lens.

The deviation produced by the lens is,

$$\delta = (\mu - 1)A \rightarrow (i)$$

Now in $\triangle BOF$, we have,

$$\tan \delta = \frac{OB}{OF} = \frac{h}{f}$$

since ' δ ' is very small, $\tan \delta \approx \delta$

$$\therefore \delta = \frac{h}{f} \rightarrow (ii)$$

From eqn (i) & (ii) we get,

$$\frac{h}{f} = (\mu - 1)A$$

$$\therefore \frac{1}{f} = (\mu - 1) \frac{A}{h} \rightarrow (iii)$$

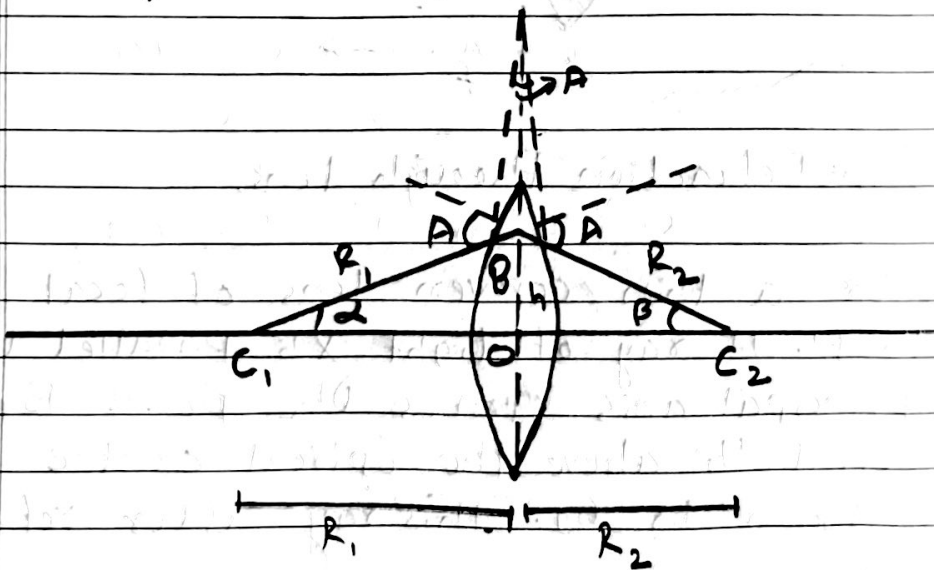


fig- lens as a small angled prism

consider two light rays passing through C_1 and C_2 meet at point B of the lens. Let $OC_1 = R_1$ and $OC_2 = R_2$ be radii of curvature of the lens and ' α ' and ' β ' are the angles made by the normal BC_1 and BC_2 with the principal axis.

By geometry,

$$A = \alpha + \beta \rightarrow (iv)$$

Also,

$$\tan \alpha = \frac{h}{R_1} \quad \text{and} \quad \tan \beta = \frac{h}{R_2}$$

But ' α ' and ' β ' are very small. So, we can write,

$$\alpha = \frac{h}{R_1} \quad \text{and} \quad \beta = \frac{h}{R_2}$$

Using these values in eqn (iv), we get.

$$A = \frac{h}{R_1} + \frac{h}{R_2}$$

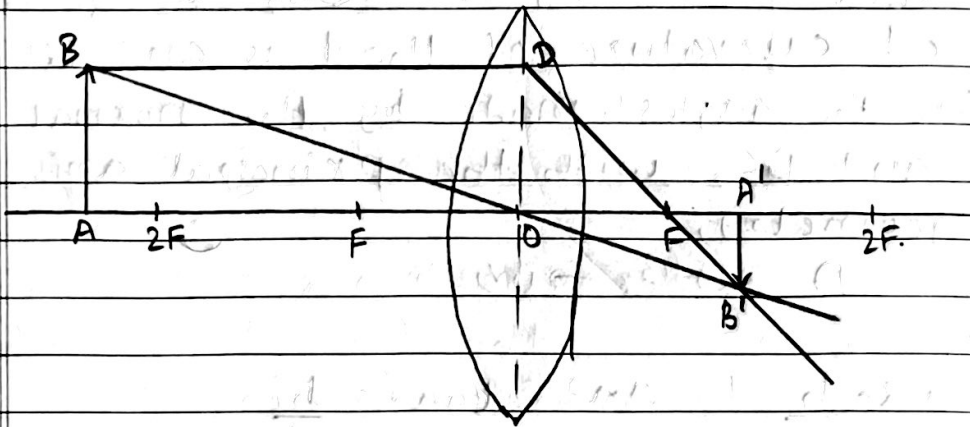
$$\frac{A}{h} = \frac{1}{R_1} + \frac{1}{R_2} \rightarrow (v)$$

From eqn (iii) and (iv)

$$\therefore \frac{1}{f} = (n-1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad \text{This is the required lens maker's formula.}$$

Lens Formula

① For convex lens when real image is formed



Consider a convex lens of focal length 'f'. ~~for let AB is~~ and optical centre O. Object AB is beyond 2F. The refracted light meet at another side of lens and form real image A'B' as shown in figure.

$$\triangle ABO \sim \triangle A'B'O$$

$$\frac{AB}{A'B'} = \frac{OA}{OA'} \rightarrow \textcircled{i}$$

$$\text{Also, } \triangle ODF \sim \triangle A'B'F$$

$$\frac{OD}{A'B'} = \frac{OF}{FA'}$$

$$\frac{AB}{A'B'} = \frac{OF}{A'F} \rightarrow \textcircled{ii}$$

From $\textcircled{i}$ & $\textcircled{ii}$

$$\frac{OA}{OA'} = \frac{OF}{A'F}$$

$$\frac{OA}{OA'} = \frac{OF}{OA' - OF} \rightarrow \textcircled{iii}$$

From sign convention

$$OA = u$$

$$OA' = v$$

$$OF = f$$

from $\textcircled{iii}$

$$\frac{u}{v} = \frac{f}{v - f}$$

$$uv - uf = vf$$

$$uv = vf + uf$$

Divide both side by uvf we get.

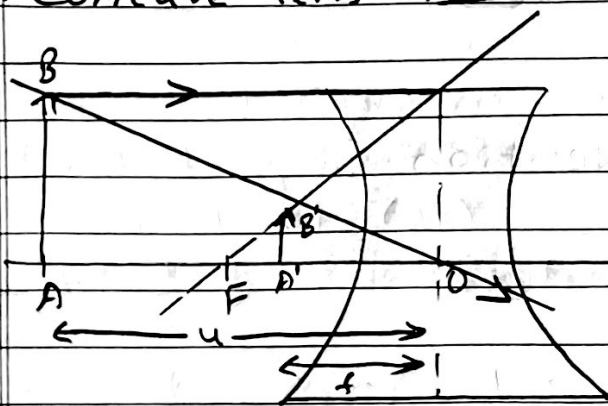
$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

from $\textcircled{i}$ $\frac{AB}{A'B'} = \frac{OP}{OA'}$

$$\frac{o}{I} = \frac{u}{v}$$

$$\Rightarrow \frac{I}{o} = \frac{v}{u} = m$$

∴

III. Concave lens

Consider a concave lens of focal length f and optical centre O . AB is object and ray of light converge at the same side of lens at a point to give a image $A'B'$

$$\triangle ABO \sim \triangle A'B'O$$

$$\therefore \frac{AB}{A'B'} = \frac{OA}{OA'} \rightarrow \textcircled{i}$$

$$\text{Also, } \triangle ABO \sim \triangle A'B'F$$

$$\therefore \frac{OA}{A'B'} = \frac{OF}{A'F}$$

$$\text{or, } \frac{AB}{A'B'} = \frac{OF}{A'F}$$

$$\text{or, } \frac{AB}{A'B'} = \frac{OF}{A'F} \rightarrow \textcircled{ii}$$

From $\textcircled{i}$ and $\textcircled{ii}$

$$\frac{OA}{OA'} = \frac{OF}{A'F}$$

$$\frac{OA}{OA'} = \frac{OF}{OF - OA'}$$

From sign convention

$$OA = u$$

$$OA' = -v$$

$$OF = -f$$

$$\frac{u}{-v} = \frac{-f}{-f + v}$$

$$-uf + uv = vf$$

$$uv = vf + uf$$

Divide both side by uvf

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Also from (i) $\frac{AB}{A'B'} = \frac{OA}{OA'}$

$$= \frac{0}{+I} = \frac{u}{+v}$$

$$\frac{I}{O} = \frac{v}{u} = m$$

• Power of Lens

→ It is defined as the reciprocal of focal length.

$$\text{i.e. Power (P)} = \frac{1}{\text{focal length (f) [In meters]}}$$

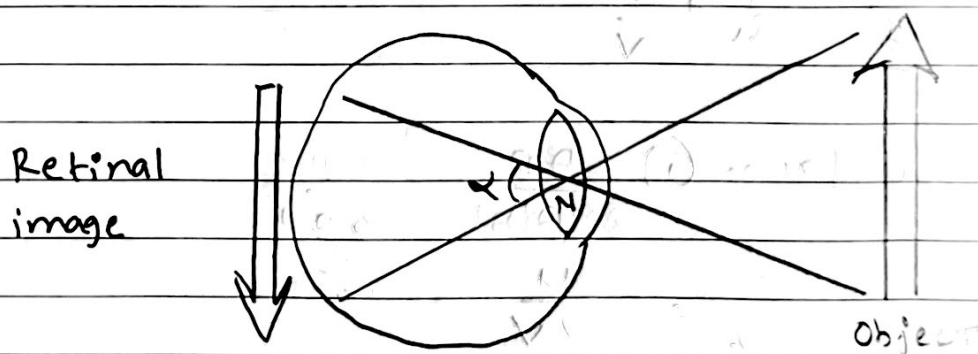
> +ve for convex lens

> -ve for concave lens

> Unit : Dioptre (D)

• Visual angle

→ The angle subtended by an object or image on our eye is known as Visual angle.



N = nodal point

α = Angle of nodal point.

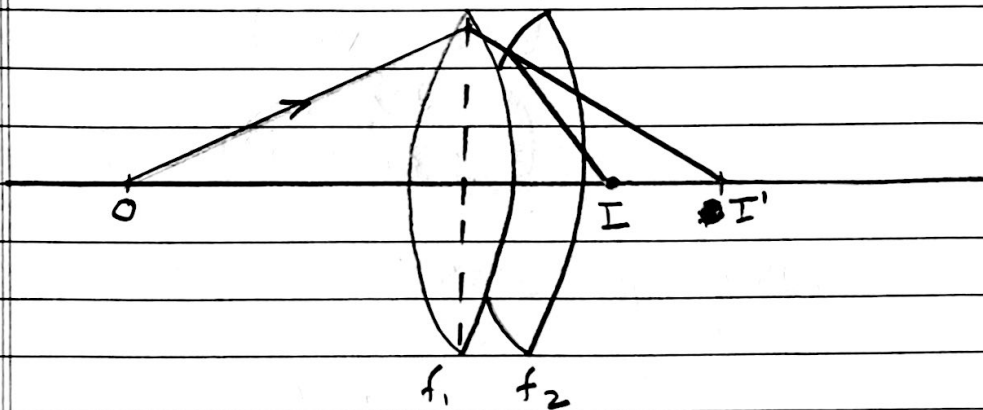
• Magnifying power

→ Magnifying power of an optical instrument is defined as the ratio of the visual angle subtended by the image at the eye (β) to the visual angle sub-

tended by the object at the unaided eye (α).

$$M = \frac{\beta}{\alpha}$$

combined focal length of two thin lens in contact



Let 'F' be the combined focal length of lens l_1 & l_2 then,

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$f = \infty$$

$$P = P_1 + P_2$$

Formula:

$$(1) \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$(2) \mu = \frac{\sin i}{\sin r}$$

$$(3) m = \frac{v}{u} = \frac{I}{O}$$

$$(4) m = -ve \text{ [erect and virtual]} \\ = +ve \text{ [inverted and real]}$$

$$(5) \boxed{P = \frac{1}{f}}$$

$$(6) \frac{1}{f} = (21-1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

Qn 5) An equiconvex lens has a power of 4 dioptre. What will be the radius of curvature of each surface if the lens is made of glass of $\mu = 1.5$?

Soln,

For equiconvex lens

$$R_1 = R_2 = R \text{ (let)} = ?$$

$$\mu_{lg} = 1.5$$

$$P = 4D$$

$$\Rightarrow f = \frac{1}{P} = \frac{1}{4} = 0.25 \text{ m}$$

$$\therefore f = 25 \text{ cm}$$

Now,

$$\frac{1}{f} = (\mu_{lg} - 1) \left(\frac{1}{R} + \frac{1}{R} \right)$$

$$\frac{1}{f} = (1.5 - 1) \times \frac{2}{R}$$

$$\Rightarrow R = 0.5 \times 2 \times f = 0.5 \times 2 \times 25 \text{ cm} \\ = 25 \text{ cm}$$

6) A convex lens of focal length 24 cm and of refractive index 1.5 is totally immersed in water of refractive index 1.33. Find its focal length in water.

→ Soln,

Air glass

Water-glass

$$f_a = 24 \text{ cm}$$

$$\mu_{lg} = 1.5$$

$$f_w = ?$$

$$\mu_{lw} = 1.33$$

$$\mu_{lg} = \frac{\mu_{lg}}{\mu_{lw}} = \frac{1.5}{1.33} = 1.127$$

We have,

$$\frac{1}{f_a} = (\mu_g - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \rightarrow \textcircled{i}$$

$$\frac{1}{f_w} = (\mu_g - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \rightarrow \textcircled{ii}$$

From eqn $\textcircled{i}$ and $\textcircled{ii}$

$$\frac{\frac{1}{f_a}}{\frac{1}{f_w}} = \frac{(\mu_g - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)}{(\mu_g - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)}$$

$$\Rightarrow \frac{f_w}{f_a} = \frac{1.5 - 1}{1.127 - 1}$$

$$\Rightarrow f_w = \frac{0.5}{0.127} \times f_a = 3.937 \times 24 = 94$$

7) The radii of curvature of the faces of a thin converging meniscus lens of glass of refractive index 1.5 are 15 cm and 30 cm. What is the focal length of the lens when it is completely immersed in water of refractive index $\frac{4}{3}$?

→ Soln,

$$R_1 = +15, R_2 = -30 \text{ cm}$$

$$\frac{1}{f_w} = (\mu_g - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$= (1.127 - 1) \left(\frac{1}{15} - \frac{1}{30} \right)$$

$$\therefore f_w = 234.7 \text{ cm}$$

$$= 23.4 \text{ m}$$

$$* \mu_{wg} = \frac{4}{3}, \mu_{lg} = 1.5$$

$$\mu_{lg} = \frac{\mu_{lg}}{\mu_{wg}}$$

$$= \frac{1.5}{1.33}$$

$$= 1.127$$

8)

a) Can a concave and Convex lens be combined so that the combination behaves like a diverging lens? Justify your answer.

→ Yes, a concave and convex can be combined so that the combination behaves like diverging lens. For that, the power of concave lens (i.e. focal length of concave lens be smaller than that of convex lens).

b) A converging meniscus of glass ($\mu = 1.5$) having radius of curvature 4 cm and 8 cm is put on a horizontal surface facing upward. If it is filled with water, what will be the focal length of the combination?

→ Soln,

$$\frac{1}{f} = \frac{1}{f_w} + \frac{1}{f_g}$$

Glass lens

$$a_{lg} = 1.5, R_1 = 4 \text{ cm}, R_2 = -8 \text{ cm}$$

$$\frac{1}{f_g} = (a_{lg} - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f_g} = (1.5 - 1) \left(\frac{1}{4} - \frac{1}{8} \right)$$

$$\therefore f_g = 16 \text{ cm}$$

Water lens

$$R_1 = +8 \text{ cm}, R_2 = \infty, a_{lw} = 1.33$$

$$\frac{1}{f_w} = (a_{lw} - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f_w} = (1.33 - 1) \left(\frac{1}{8} + \frac{1}{\infty} \right)$$

$$\therefore f_w = 24 \text{ cm}$$

Now,

$$\frac{1}{f} = \frac{1}{16} + \frac{1}{24}$$

$$\therefore f = 9 \text{ cm}$$

4) The image obtained with a converging lens is erect and three times the length of object. The focal length of the lens is 20 cm. Calculate the image distance.

→ Given,

~~Concave~~ Converging lens

Object distance = 0

image size = $I = -30$

focal length = 20 cm

Now,

$$M = \frac{I}{O} = \frac{V}{u}$$

$$\text{or, } \frac{-30}{O} = \frac{V}{u}$$

$$\text{or, } V = -3u$$

Now,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{V}$$

$$\text{or, } \frac{1}{20} = \frac{1}{u} - \frac{1}{3u}$$

$$\text{or, } 20 = \frac{3-1}{3u}$$

$$\text{or, } 20 = \frac{3u}{2}$$

$$\text{or, } u = \frac{40}{3}$$

Again,

$$V = -3u = -3 \times \frac{40}{3} = -40 \text{ cm}$$

Hence, image distance is -40 cm

3) The real image obtained by a lens of power 50 is three times the length of object. Calculate the object and image distance.

→ Given,

$$P = 50$$

$$\text{or, } \frac{1}{f} = 50$$

$$\text{or, } f = \frac{1}{5} \text{ m}$$

$$\text{or, } f = \frac{1}{5} \times 100 \text{ cm}$$

$$\therefore f = 20 \text{ cm}$$

Conver lens

Size of image (I) = 3 x size of object (O)

$$\text{or, } I = 3O$$

$$M = \frac{I}{O} = \frac{4}{V}$$

$$\text{or, } \frac{3O}{O} = \frac{V}{4}$$

$$\text{or, } V = 3u$$

Now,

$$\frac{1}{f} = \frac{1}{V} + \frac{1}{u}$$

$$\text{or, } \frac{1}{f} = \frac{1}{3u} + \frac{1}{u}$$

$$\text{or, } \frac{1}{20} = \frac{1+3}{3u}$$

$$\text{or, } 20 = \frac{3u}{4}$$

$$\text{or, } u = \frac{80}{3} \text{ cm}$$

$$\therefore u = 26.67 \text{ cm}$$

$$\therefore V = 3u = 3 \times \frac{80}{3} = 80 \text{ cm}$$

2) An erect object is placed at a distance of 10 cm before a convex lens of focal length 15 cm. When will the image be formed? Discuss the nature of image.

↳ Given,

Object distance (u) = 10 cm

Focal length (f) = 15 cm

Since Object distance is lesser than focal length so, object is between O and F and Virtual image will be formed.

∴ image distance = v

Now,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\text{or, } \frac{1}{15} = \frac{1}{10} + \frac{1}{v}$$

$$\text{or, } \frac{1}{15} = \frac{v + 10}{10v}$$

$$\text{or, } 10v = 15v + 150$$

$$\text{or, } 5v = -150$$

$$\text{or, } v = -30 \text{ cm}$$

∴ Image is virtual,

$$\frac{v}{u} = \frac{30}{10} = 3$$

Hence image is 3 times magnified.

1) The given figure shows a ray of light is entering and emerging through a part of conver lens.

a) Define conver lens and state one daily application of it.

b) Explain why this lens is also called converging lens?

c) Calculate the refractive index of material of lens shown in figures.

Ans Conver lens is a lens which is thick at the middle and thin at the edges, and it converges a parallel beam of light falling on it to a point.

one Application of conver lens is;

- Used in eye glasses for hypermetropia.

Ans This lens is also called converging lens because it converges a parallel beam of light falling on it to a point.

c) Given

$$i = 54^\circ, r = 32^\circ$$

$$\mu = \frac{\sin i}{\sin r}$$

$$= \frac{\sin 54^\circ}{\sin 32^\circ}$$

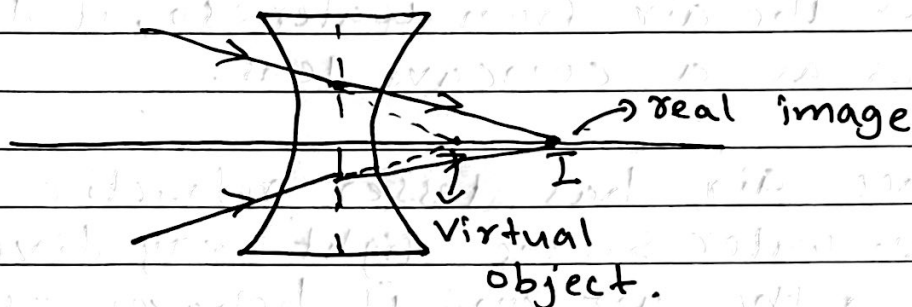
$$\therefore \mu = 1.53$$

Hence, the refractive index is 1.53.

• Conceptual (Short) Questions.

1) Can a Concave lens form a real image?
Give the condition.

Ans Yes a concave lens can sometimes form a real image. When the object is virtual, the concave lens can form a real image if, a converging ray of light is incident on the concave lens.



2) ~~How~~ Does the focal length of a lens change when immersed in water? will it increase or decrease?

Ans We have,

$$\frac{1}{f} = (n_l - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

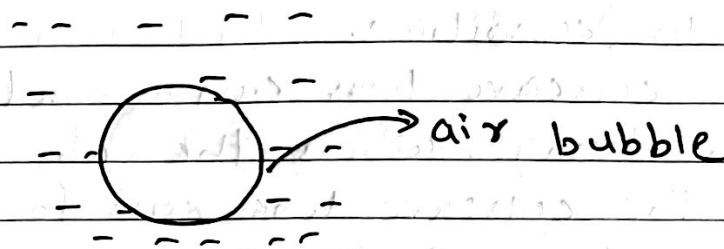
If the lens is immersed in water,

$$n_{wg} < n_{wg} \\ \therefore f_g > f_a$$

And power decreases, since $P = \frac{1}{f}$

So, focal length will increase if a lens is immersed in water.

3) Does the air bubble inside water behave as a concave or convex lens?



↳ The ray of light diverges when it strikes the air from water so, it behaves as a concave lens.

Since, air has lesser refractive index than water so, the light ray diverges from the air and it behaves as concave lens.

④ Sketch the ray diagram showing the formation of a virtual image by a convex lens.

↳

